

PetCare

Android Pet Care Management App

Software Requirements Specification - Functional Requirements

Document	Functional requirements, business rules, data model and build plan
Source	Reverse-engineered from the PetCare Android codebase (package com.example.petcare), 46 Kotlin source files, about 11,100 lines
Platform	Android 10+ (minSdk 29, targetSdk 36), Kotlin, XML layouts, Material 3
Storage	On-device SQLite (database version 14) + SharedPreferences
Date	27 September 2026
Version	1.0

This document lists every feature the current PetCare app has: what each screen does, which inputs it accepts, which validation and business rules apply, and how the parts connect. It is written so the project can be rebuilt, extended or documented (for example in a project report) from it alone.

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1. Introduction

1.1 Purpose

PetCare is a personal organiser for pet owners. It keeps each pet's profile, daily/weekly/monthly care routines, health records and vaccinations, expenses and favourite places in one app, and reminds the owner when something is due. All data lives on the phone; nothing is uploaded to a server.

1.2 Scope

The app covers these functional areas (each is a section in Chapter 3):

#	Module	Summary
A	Authentication & session	Sign up, login, Remember me, biometric unlock, Google sign-in, forgot/change password, logout
B	Home dashboard	Greeting, today's progress ring, stats, Up next, quick actions, pet cards, emergency contact, tip of the day
C	Pet profile management	Add/edit pet with photos, species, allergies, vaccination date, completion meter; delete pet
D	Pet details	Profile view, vaccination card, about rows, photo gallery, health records, shortcuts
E	Health records & vaccination	Record types, next-due logic, automatic schedule updates
F	Care routines (Tasks)	Create routines with repeat rules, reminders, supplies, notes, linked place; list, filter, edit, delete
G	Daily checklist	Per-pet or all-pets checklist, search, weekly day ticks, add to calendar, upcoming health card
H	Completion rules	When a routine may be marked done (time gate, no undo, no future days)
I	Reminders & notifications	Exact alarms for routines, vaccinations and health records; reboot recovery
J	Notifications screen	Overdue / upcoming / scheduled reminders, health history, today's log
K	Expenses	Add expense, filter by pet/period/search, monthly grouping, category breakdown, PDF/CSV export
L	Places (Locations)	Save vet clinics, parks, shops etc. with a map pin; open directions; link to routines
M	Delegate care	Build a care summary for a pet sitter and send it by SMS
N	Import records	Import vaccinations/appointments from .ics calendar or .csv files
O	Share & deep link	Receive shared text or petcare:// links and save as a health record
P	Pet history	Read-only timeline of care activity, expenses and health records per pet
Q	Profile & settings	Profile photo, personal info, emergency phone, theme, notifications, help/privacy/terms
R	Gestures & navigation	Bottom navigation, context FAB, swipe actions, long-press menus, shake to open checklist

1.3 Users

- **Pet owner** (primary user): registers, manages pets, routines, expenses and records.
- **Delegate / pet sitter** (indirect user): does not use the app, but receives the care summary SMS and can be called from the emergency contact card.

1.4 Technology stack

Area	Used in the project
Language / UI	Kotlin, Android Views with XML layouts, Material Components 3 (chips, cards, dialogs, snackbars), custom Figtree font, light & dark themes
Database	SQLiteOpenHelper (AuthDatabaseHelper), 8 tables, schema version 14 with incremental migrations
Preferences	SharedPreferences "PetCarePrefs" (session, theme, notification toggle, profile photo, emergency contact)
Security	SHA-256 password hashing salted with the normalised email; androidx.biometric for fingerprint/face/device PIN
Google sign-in	androidx.credentials Credential Manager + Google ID token (googleid library)
Maps	osmdroid 6.1.20 with OpenStreetMap tiles; Android Geocoder with Nominatim (OpenStreetMap) fallback
Background	AlarmManager exact alarms + BroadcastReceiver; single-thread executors for database work off the UI thread
Files	MediaStore Downloads (no storage permission), Storage Access Framework file picker, PdfDocument for PDF export
Sensors	Accelerometer (shake detection)
Tests	JUnit unit tests: RecordImporterTest (7 tests), TaskScheduleTest (6 tests)

2. System overview

2.1 Screens

Screen (Activity)	Opened from	Purpose
MainActivity	App launcher	Welcome, Login and Sign-up panels; skips to Dashboard if already logged in
ForgotPasswordActivity	Login > Forgot password	Reset a password by email
DashboardActivity	After login, notification tap	Hosts 4 bottom tabs: Home, Tasks, Expenses, Profile
AddEditPetActivity	FAB, Add pet, pet details > Edit	Create or edit a pet profile
PetDetailsActivity	Tap a pet card	Full pet profile, vaccination, photos, health records
AddTaskActivity	Tasks FAB, quick action, pet details	Create a care routine
ChecklistActivity	Up next, task tap, shake, pet details	Today's checklist for one or all pets
NotificationsActivity	Bell icon on Home/Tasks	Reminder centre
AddExpenseActivity	Expenses FAB, quick action, pet details	Record an expense
LocationsActivity	Profile > Locations, quick action Places	Saved places list
LocationMapActivity	Locations (pick or view)	Map picker / map viewer with directions
DelegateContactActivity	Quick action Delegate, Profile	Send care details by SMS
ImportRecordsActivity	Profile > Import records; opening/sharing an .ics/.csv	Import health records from files
ShareReceiverActivity	Share text to PetCare; petcare:// link	Save a shared appointment as a record
PetHistoryActivity	Profile > Pet history	Read-only history per pet
PersonalInfoActivity	Profile > Personal info, emergency card	Edit name, emergency phone, photo
ChangePasswordActivity	Profile > Change password	Set a new password
InfoActivity	Profile > Help / Privacy / Terms	Help Center FAQ and legal pages

2.2 Main user flow

- Launch -> Welcome -> Create account or Log in (password, Google or biometric) -> Dashboard (Home tab).
- Home -> Add pet -> the pet gets 4 starter routines automatically -> Tasks/Checklist to tick them off during the day.
- Reminders fire at each routine's time; tapping a notification opens the Tasks tab.
- Expenses tab to record spending and download reports; Profile tab for places, imports, history, settings and logout.

3. Functional requirements

Each requirement has an ID (FR-<module>-<number>), a short statement, and the exact behaviour and validation implemented in the code. Messages in quotes are the texts the app shows.

A. Authentication & session

ID	Requirement	How it works / rules
FR-AUTH-01	Welcome screen	First screen for a logged-out user with two buttons: <i>Create account</i> and <i>Login</i> . Device Back on Login/Sign-up returns to Welcome; forms are cleared when switching panels.
FR-AUTH-02	Auto login	If a session exists (is_logged_in = true in preferences) the app opens the Dashboard directly.
FR-AUTH-03	Sign up	Fields: full name, email, password, confirm password, <i>accept Terms</i> checkbox. Rules, in order: all fields required; valid email format; password at least 6 characters; passwords match; terms accepted; email not already registered. On success: "Account created. You can login now." and the Login panel opens with the email pre-filled.
FR-AUTH-04	Password strength meter	4-segment live meter on sign-up: empty = grey; under 6 chars = Weak (1 red segment); 6-7 chars = Medium (3 segments); 8+ = Strong (4 green segments).
FR-AUTH-05	Show/hide password	Eye icon on every password field toggles visibility.
FR-AUTH-06	Login	Email + password; both required. The password is hashed and compared with the stored hash. Wrong credentials: "Invalid email or password.", password field cleared, autofill session cancelled.
FR-AUTH-07	Remember me	If ticked, the email is stored and pre-filled next time; unticking removes it. Survives logout.
FR-AUTH-08	Biometric login	Card on the Login panel: "Sign in as <email>". Unlocks the last account that signed in on this device (or the typed email if registered). Uses fingerprint/face or device PIN (strong biometrics on Android 11+, weak on Android 10). Shows clear messages when nothing is enrolled or the sensor is unavailable.
FR-AUTH-09	Continue with Google	Uses Credential Manager to get a Google ID token. First use creates a local account (no password, so it signs in only via Google or biometrics); later uses reuse the account with the same email. Requires google_web_client_id in strings.xml, otherwise shows a configuration message.
FR-AUTH-10	Forgot password	Email + new password + confirm. Rules: all fields required, 6+ chars, match, email must exist. Updates the stored hash and returns to Login.
FR-AUTH-11	Change password	Logged-in user enters new + confirm password (6+ chars, must match). After saving, the user is logged out and must log in again.
FR-AUTH-12	Logout	Profile > Logout asks for confirmation, clears the session but keeps the Remember-me and biometric email, and returns to the Welcome screen.
FR-AUTH-13	Data isolation per user	Pets carry an owner_id; every pet, task, expense and record query is filtered to the logged-in user. Pets without an owner are adopted by the current user.

B. Home dashboard

ID	Requirement	How it works / rules
FR-HOME-01	Greeting header	"Good morning / afternoon / evening / night - Sat, 27 Sep" based on the hour (5-11, 12-16, 17-20, else), the user's name, and the profile photo or the user's initials.

ID	Requirement	How it works / rules
FR-HOME-02	Today card	Circular progress ring with % of today's due routines done, "X of Y done", and a summary line: add-a-pet prompt, add-a-routine prompt, "All routines complete", or "Next/Overdue: <task> - <pet> at <time>". Tapping opens the Tasks tab.
FR-HOME-03	Stat tiles	Pets count (tap = add pet); Due today count with "N overdue" in red when times have passed (tap = Tasks); Spent this month in \$ (tap = Expenses).
FR-HOME-04	Up next list	The next 3 pending routines sorted by time with time pill (red if overdue), pet and category. Tapping opens that pet's checklist. Empty text: "No routines scheduled for today." or "You're all caught up".
FR-HOME-05	Quick actions	Five shortcuts: Add pet, Add task, Expense, Delegate, Places.
FR-HOME-06	My pets list	One card per pet: photo (or paw icon), name, breed - species, status line ("N care tasks pending" in red / "All care tasks done" / upcoming vaccine), progress bar with done/total, and 4 care chips (Fed, Walked, Meds, Groomed) coloured when that category was done today. Tap opens Pet details.
FR-HOME-07	Anniversary badge	A festive badge appears on a pet card on the yearly anniversary of the day it was added.
FR-HOME-08	Emergency contact card	Shows the saved emergency phone with a Call action (opens the dialer); if none is set it shows "Not set up yet" and opens Personal info.
FR-HOME-09	Tip of the day	One of 8 care tips, changing daily (same tip all day).
FR-HOME-10	Notification bell	Visible only when notifications are enabled; a red dot shows while routines are pending. Opens the Notifications screen.
FR-HOME-11	Background loading	Pets, tasks and monthly spend load on a background thread and refresh every time the screen resumes.

C. Pet profile management

ID	Requirement	How it works / rules
FR-PET-01	Add / edit pet form	Sections: Basic (name, species, breed, age, weight), Health (diet, allergies, next vaccination date, reminder switch), Notes (favourite toys & comfort items, behaviour notes), Photos. Title and button change to "Edit pet" / "Save changes" in edit mode.
FR-PET-02	Validation	Name required. Age optional, whole number 0-40. Weight optional, > 0 and up to 200 kg. Vaccination date cannot be in the past. Errors show under the field and clear when the user types; toast "Please fix the highlighted fields".
FR-PET-03	Species	Single-choice chips: Dog, Cat, Bird, Rabbit, Other. Choosing Other shows a text field for a custom species.
FR-PET-04	Allergies	Multi-select chips (None, Chicken, Beef, Dairy, Grain, Pollen, Fleas, Penicillin) plus a free-text field for others. "None" is exclusive. Saved as one comma-separated value without duplicates.
FR-PET-05	Photos	Pick up to 5 images from the system photo picker (persistable permission). The first photo is the main avatar. Tapping a thumbnail offers <i>Make main photo</i> / <i>Remove photo</i> .
FR-PET-06	Next vaccination date	Date picker with today as the minimum date and a Clear button. Stored as dd/MM/yyyy.
FR-PET-07	Vaccination reminder switch	If on and a date is set, schedules a 9:00 AM notification on that day; otherwise cancels it.

ID	Requirement	How it works / rules
FR-PET-08	Profile completion meter	Live progress bar: 11 items (9 text fields + species + at least one photo) -> percentage.
FR-PET-09	Unsaved changes guard	Leaving with unsaved edits asks "Discard changes?" (Keep editing / Discard).
FR-PET-10	Double-save protection	Save button is disabled while saving so a double tap can't create two pets.
FR-PET-11	Starter routines	When a new pet is created, 4 daily routines are seeded once: Morning Kibble & Wet Food (Feeding 7:30 AM), Afternoon Fur Brushing (Grooming 2:00 PM), Ear Drops & Multivitamin (Health 6:00 PM), Evening Laser Chase (Activity 8:00 PM), each with supplies and notes.
FR-PET-12	Delete pet	From Pet details menu with confirmation. Deletes, in one transaction, the pet's task completions, tasks, expenses, photos and health records, and cancels its vaccination reminder.

D. Pet details

ID	Requirement	How it works / rules
FR-DET-01	Header	Avatar (tap = full-screen photo), name, species - breed, badge "X of Y routines done today".
FR-DET-02	Quick stats	Age ("1 yr"/"N yrs"), weight in kg, total spent on this pet.
FR-DET-03	Vaccination card	Shows next vaccination date with a status pill: Overdue Nd (red), Today, In N days (amber if within 14 days), In N days. Tap offers: Mark as given... / Change date / Remove date, or Set date if empty.
FR-DET-04	About rows	Diet; Allergies & medical warnings (red when real allergies exist); Favourite toys; Behaviour notes. Empty values show "Not added".
FR-DET-05	Photo gallery	Horizontal list of all photos, count in the header, tap for full screen; Manage opens the editor.
FR-DET-06	Health records list	Sorted upcoming first (soonest), then past (newest). Each row: type icon/colour, type, date, notes, "Tomorrow"/"In N days" pill for future records, delete button with confirmation (also cancels its reminder).
FR-DET-07	Shortcuts	Open checklist, Add task and Add expense for this pet (pet pre-selected).
FR-DET-08	Toolbar menu	Edit pet and Delete pet.
FR-DET-09	Auto refresh	Reloads on every return so edits, new tasks and expenses appear; closes itself if the pet no longer exists.

E. Health records & vaccination

ID	Requirement	How it works / rules
FR-HLT-01	Record types	Vaccination, Check-up, Medication, Surgery, Other. Each type has its own icon and colour.
FR-HLT-02	Add health record	Dialog: type chips, date (defaults to today, any date allowed), notes. For Vaccination an extra <i>Next due</i> date field appears (minimum tomorrow); saving it updates the pet's next vaccination date.
FR-HLT-03	Mark vaccine as given	From the vaccination card: opens the record dialog pre-set to Vaccination.
FR-HLT-04	Automatic scheduling (HealthSchedule)	After any record is saved (manual, import or share): past records are history only. A future Vaccination becomes the pet's next vaccination if it is sooner than the current one (and gets the vaccine reminder if enabled). Any other future record gets a 9:00 AM reminder on its day.

ID	Requirement	How it works / rules
FR-HLT-05	Remove next vaccination	Confirmation dialog; removes the date and its reminder, health records stay.

F. Care routines (Tasks)

ID	Requirement	How it works / rules
FR-TASK-01	Add care task form	Choose pet (dropdown; pre-selected when opened from a pet), description, category chips (Feeding, Exercise, Grooming, Medication, Healthcare, Cleaning), repeat (Daily / Weekly / Monthly), time, ends-on date, optional expense amount, required supplies, special notes, linked saved place, <i>Remind me</i> and <i>Delegate this task</i> checkboxes.
FR-TASK-02	Validation	A pet must exist and be chosen ("Add a pet first"); description required. Monthly needs a day 1-31.
FR-TASK-03	Repeat rules	Daily = every day. Weekly = on selected weekdays (Mon-Sun toggles, at least one always selected, stored like "Mon,Wed,Fri"). Monthly = one day of the month (defaults to today's day); in shorter months a 29-31 routine falls on the last day.
FR-TASK-04	Time picker	12-hour picker; stored as "hh:mm AM/PM". A routine without time is "Anytime".
FR-TASK-05	Linked place	Dropdown of saved locations ("None / No Location" first). If none are saved, tapping explains how to add one.
FR-TASK-06	Reminder on save	If Remind me is ticked and a time is set, the next occurrence is scheduled immediately.
FR-TASK-07	Tasks page - date	Header date chip ("Today" or "Sep 30") opens a date picker; the list shows routines due on that day.
FR-TASK-08	Tasks page - filters	Pet chips (All pets + each pet) and category chips All / Feeding / Grooming / Health / Activity, each showing its count ("Feeding - 2"). Category matching uses keywords (e.g. Health = health, med, vet).
FR-TASK-09	Tasks page - progress	"X of Y completed", bar with %, hint: "N tasks left - Next: ..." or "All done for the day".
FR-TASK-10	Task row	Coloured accent bar and category badge, pet badge, time plus supplies/notes/repeat label; completed rows are dimmed, struck through, badge "Done".
FR-TASK-11	Row actions	Tap = open checklist; tap circle = complete; swipe right = complete; swipe left = delete (with confirmation); 3-dot menu or long-press = Mark complete / Open checklist / Edit / Delete.
FR-TASK-12	Edit routine	Dialog to change title, time, supplies and notes (title required). The reminder moves to the new time. Completed routines can't be edited.
FR-TASK-13	Delete routine	Confirmation; deletes the routine and its completion history and cancels its reminder.

G. Daily checklist

ID	Requirement	How it works / rules
FR-CHK-01	Scope	Shows today's due routines for one pet or "All pets checklist"; pet chips at the top switch between them.
FR-CHK-02	Progress	"X of Y done today" / "All N done today" with bar and %.
FR-CHK-03	Search	Live search (250 ms debounce) over title, category, pet name, notes and supplies.

ID	Requirement	How it works / rules
FR-CHK-04	Sorting	Open routines first, then by time, then alphabetically.
FR-CHK-05	Row content	Title, time - repeat label - pet name (when mixing pets), category pill, supplies and notes. Before its time the circle is dimmed and tapping explains when it unlocks.
FR-CHK-06	Weekly day ticks	Weekly routines show one chip per chosen day with "This week - X of N days". A day can be ticked only when it has arrived (and its time has passed today); ticked days can't be unticked.
FR-CHK-07	Add to calendar	Opens the phone's calendar app with a 30-minute event "PetCare: <task> (<pet>)" at the routine's next time, with supplies, notes and linked place as location.
FR-CHK-08	Upcoming health card	Shows the nearest health record within the next 30 days ("Bella's vaccination in 3 days", +N more); tap opens that pet.
FR-CHK-09	Edit / delete	Tap a row or use the menu to edit (title required); delete with confirmation.
FR-CHK-10	Optimistic UI	Changes appear instantly and are rolled back with a message if the database write fails.
FR-CHK-11	Add button	FAB opens Add task for the selected pet; it hides while scrolling down.

H. Completion rules (apply on every screen)

ID	Requirement	How it works / rules
FR-RULE-01	Done is final	A completed routine cannot be reopened: "Already done. Completed routines can't be undone."
FR-RULE-02	Time gate	A timed routine unlocks at its time: "Available from 02:00 PM. You can mark it done once it's time." Routines without a time can be done any time that day.
FR-RULE-03	No future days	"You can't complete routines for a future day." Past days selected on the Tasks page can be completed.
FR-RULE-04	Per-day completion	Completion is stored per task per date (task_completions table), so every day starts fresh.
FR-RULE-05	Weekly days	Same rules per weekday (Mon-Sun of the current week).

I. Reminders & notifications

ID	Requirement	How it works / rules
FR-REM-01	Notification channel	"Care Reminders" (high importance) created at app start.
FR-REM-02	Routine reminders	One exact alarm per routine for its next due time (looks up to 63 days ahead). When it fires and the routine is due and not done, a notification is shown and the next occurrence is scheduled (daily -> tomorrow, weekly -> next chosen day, monthly -> next month).
FR-REM-03	Notification titles	By category: Meal time, Walk time, Medication reminder, Grooming time, Cleaning time, Health check, or Care reminder. Text: "<pet>: <task>".
FR-REM-04	Vaccination reminder	9:00 AM on the due date: "Vaccination due today".
FR-REM-05	Health record reminder	9:00 AM on the record's date: "<Type> today - <pet> has a <type> scheduled for today".
FR-REM-06	Tap action	Notification opens the Dashboard on the Tasks tab.
FR-REM-07	Global switch	Profile > Notifications toggle; when off, alarms still run but show nothing and the bell is hidden.

ID	Requirement	How it works / rules
FR-REM-08	Reboot / update recovery	On BOOT_COMPLETED or app update every routine, vaccination and record alarm is re-created.
FR-REM-09	Exact alarm fallback	If exact alarms aren't allowed, an inexact alarm is used instead.

J. Notifications screen

ID	Requirement	How it works / rules
FR-NOTI-01	Today's reminders	Pending routines due today grouped by state: Overdue (red card, first), Upcoming ("in N minutes", within 60 min), Scheduled. Tap opens the pet checklist.
FR-NOTI-02	Healthcare history	All health records with pet, type, date and a Notes tag.
FR-NOTI-03	Log section	Today's due routines with Done / Missed status (labelled "Yesterday" in the layout).
FR-NOTI-04	Bottom navigation	Same four tabs as the Dashboard.

K. Expenses

ID	Requirement	How it works / rules
FR-EXP-01	Add expense	Pet (dropdown), category chips Food / Vet / Medication / Grooming / Toys, description, amount, date (defaults to today). Rules: pet required, description required, amount > 0. Save disabled when there are no pets.
FR-EXP-02	Filters	Period chips: All time, This month, Last 30 days, This year. Pet chips: All pets + each pet. Live search over description, pet and category.
FR-EXP-03	Summary	Total (label changes with period, e.g. "Spent this month"), number of transactions, average per transaction, top category, subtitle "<pet> - <period>".
FR-EXP-04	Category breakdown	Stacked colour bar and legend with amount and % per category.
FR-EXP-05	List	Newest first, grouped under month headers with the month total ("September 2026 - \$124.50"). Row: category icon, description, pet - category - date, amount.
FR-EXP-06	Delete	Swipe left on a transaction -> confirmation dialog with name and amount; cancel slides it back.
FR-EXP-07	Export	Download icon -> PDF report or CSV spreadsheet of the currently filtered list, saved to Downloads/PetCare/PetCare_expenses_<date_time>; snackbar with Open (or Share if no viewer app).
FR-EXP-08	PDF report content	Title, filter scope, generation time, total, count, per-category totals with %, table Date / Pet / Category / Description / Amount, multi-page with page numbers, grand total.
FR-EXP-09	CSV content	UTF-8 with BOM (opens correctly in Excel), header Date,Pet,Category,Description,Amount, ISO dates, total row. Cells starting with = + - @ are prefixed to prevent CSV formula injection.
FR-EXP-10	Currency format	US format \$#,##0.00; dates shown as "19 Sep 2026" and stored as dd/MM/yyyy.

L. Places (Locations & map)

ID	Requirement	How it works / rules
FR-LOC-01	Place categories	Vet clinic, Grooming salon, Dog park, Pet store, Animal shelter, Other (each with icon and colour). Old free-text categories are mapped by keyword.
FR-LOC-02	List & filter	Cards with name, category badge, address, <i>Locate on Map</i> and <i>Delete</i> . Filter chips All + each category.
FR-LOC-03	Add place	Dialog: name, type of place, address, <i>Set Location Pin on Map</i> . Name and address required.
FR-LOC-04	Map picker	OpenStreetMap map (default centre Kathmandu) with a centre pin. Moving the map reverse-geocodes the centre (Android Geocoder, then Nominatim, else raw coordinates) and shows the address and Lat/Lng. Search box forward-geocodes an address. My-location button asks for location permission and jumps to the last known position. <i>Confirm & Set Location</i> returns the coordinates and address to the form.
FR-LOC-05	Auto geocoding	If saved without a pin, the address is geocoded in the background and coordinates are stored.
FR-LOC-06	View on map	Shows a marker with name and address; <i>Get Directions</i> opens Google Maps (or any map app).
FR-LOC-07	Delete place	Confirmation dialog, then removed.
FR-LOC-08	Use in routines	Saved places can be linked to a routine and appear as the event location in calendar export.

M. Delegate care (send to a pet sitter)

ID	Requirement	How it works / rules
FR-DEL-01	Choose pet and contact	Select pet dialog; contact chosen from the phone book (name + number). The saved emergency phone is pre-filled as "Default Delegate Contact".
FR-DEL-02	What to send	One of: Checklist (pending routines with time, supplies, notes), Feeding Schedule, Medication, Emergency Summary (species, breed, age, weight, allergies, vaccine date).
FR-DEL-03	Extras	<i>Include care notes</i> switch appends diet and behaviour notes; a custom instructions box adds a personal message.
FR-DEL-04	Live preview	Message preview updates on every change and ends with "Regards, <user> (Sent via PetCare)".
FR-DEL-05	Send via SMS	Opens the SMS app with number and text filled in. Requires a pet and a recipient.

N. Import records (.ics / .csv)

ID	Requirement	How it works / rules
FR-IMP-01	Entry points	Profile > Import records (file picker), or opening/sharing an .ics or .csv file with PetCare from another app.
FR-IMP-02	iCalendar parsing	Reads each VEVENT: DTSTART (all-day, local or UTC time), SUMMARY, LOCATION, DESCRIPTION; handles folded lines and escapes. Notes = summary - time - location - description (max 300 chars).
FR-IMP-03	CSV parsing	Optional header; columns found by name (date; type/category/title/summary/event; note/description/detail/comment) or by position 1-2-3. Quoted cells supported. Date formats: yyyy-MM-dd, dd/MM/yyyy, d/M/yyyy, dd-MM-yyyy, dd.MM.yyyy. Rows without a valid date are counted as skipped.

ID	Requirement	How it works / rules
FR-IMP-04	Type detection	Keywords map text to a type: vaccin/rabies/booster/dhpp/parvo/distemper/bordetella/shot = Vaccination; check/exam/consult/visit/wellness = Check-up; surg/spay/neuter/operation/dental cleaning = Surgery; medic/deworm/tablet/pill/dose/flea/tick = Medication; else Other.
FR-IMP-05	Preview	"file.ics - N records found - M skipped", pet chips to choose the pet, a checkbox per record. Records the pet already has (same type, date and notes) are shown as "Already added" and disabled.
FR-IMP-06	Import	Button "Import N records" saves the selected ones, runs automatic scheduling (section E) and opens the pet's profile.
FR-IMP-07	Errors	Unreadable file, no dated records (with a format hint), or no pets yet. Files are capped at 1 MB.

O. Share receiver & deep link

ID	Requirement	How it works / rules
FR-SHR-01	Share text to PetCare	Text shared from SMS, email or another app opens a "Save to PetCare" dialog.
FR-SHR-02	Deep link	petcare://import?desc=...&date=2026-10-12&time=09:30 AM opens the same dialog.
FR-SHR-03	Auto detection	Detects a time (10:00 AM, 2:30pm, 14:00), a date (2026-10-12 or 12/10/2026) and the record type from the text.
FR-SHR-04	Confirm and save	User picks the pet, corrects type/date/notes and saves as a health record; automatic scheduling applies. If there are no pets: "Add a pet in PetCare first".

P. Pet history (read-only)

ID	Requirement	How it works / rules
FR-HIS-01	Pet switcher	Chips for each pet; the first pet is selected by default.
FR-HIS-02	Overview	Photo, name, species - breed - "With you since <date>"; totals: routines done, total spent, records.
FR-HIS-03	Facts	Age, weight, diet, allergies (red if set), next vaccination ("Overdue" in red / Today / in N d), active routines.
FR-HIS-04	Care activity	Completed routines grouped by day with "N done"; summary "N done in the last 30 days - M active days".
FR-HIS-05	Expenses	All expenses for the pet, newest first; summary with count and this month's total.
FR-HIS-06	Health records	Upcoming first then past; "Upcoming" tag.
FR-HIS-07	Collapsed lists	Lists longer than 5 rows show "Show all (N)" / "Show less". Nothing can be edited here.

Q. Profile & settings

ID	Requirement	How it works / rules
FR-PRO-01	Account card	Profile photo (tap to change via photo picker), name and email.

ID	Requirement	How it works / rules
FR-PRO-02	Personal info	Edit name (required) and emergency / delegate phone number; change photo. Email is shown.
FR-PRO-03	Preferences	Notifications on/off switch; App theme dialog Light / Dark (applied immediately and at every start).
FR-PRO-04	Menu items	Pet history, Import records, Personal info, Delegate contact, Locations, Change password, Help Center, Privacy Policy, Terms and Conditions, Logout.
FR-PRO-05	Help Center	12 expandable FAQ items (adding pets, routines, completion, gestures, places, import, expenses, reminders, deleting, backups) and <i>Contact support</i> that opens an email with app version and device details.
FR-PRO-06	Privacy Policy & Terms	Numbered in-app legal pages with last-updated date; app version shown at the bottom.

R. Navigation & gestures

ID	Requirement	How it works / rules
FR-NAV-01	Bottom navigation	Home, Tasks, Expenses, Profile inside one Dashboard; pages are created lazily and refreshed on selection.
FR-NAV-02	Context FAB	Home/Profile = Add pet; Tasks = Add task; Expenses = Add expense (pet filter pre-selected).
FR-NAV-03	Shake to open checklist	Shaking the phone (above 2.7 g, max once per 2 s) on the Dashboard gives haptic feedback and opens today's checklist.
FR-NAV-04	Swipes and long-press	Tasks: right = done, left = delete. Expenses: left = delete. Long-press task rows for quick actions.

4. Data model

4.1 SQLite tables

Table	Columns	Notes
users	id, name, email (unique), password_hash, created_at	Hash = SHA-256(lowercase email + ":" + password)
pets	id, name, species, breed, age, weight, diet, vaccine_date, reminder_enabled, allergies, toys, notes, owner_id, created_at	owner_id -> users.id
pet_photos	id, pet_id, uri	Content URIs of chosen photos, in display order
tasks	id, pet_id, description, category, repeat_type, week_days, scheduled_time, ends_on, delegate_enabled, is_completed, expense_amount, completed_week_days, reminder_enabled, required_supplies, task_notes, linked_location_id	week_days = weekday list or month day; is_completed mirrors "done today"
task_completions	task_id, completed_date (yyyy-MM-dd)	Primary key (task_id, completed_date)
expenses	id, pet_id, category, description, expense_date (dd/MM/yyyy), amount, created_at	
healthcare_history	id, pet_id, type, date (dd/MM/yyyy), notes	Health records
pet_locations	id, user_id, name, address, category, latitude, longitude	Saved places per user

4.2 Preferences (SharedPreferences "PetCarePrefs")

user_name, user_email, is_logged_in, notifications_enabled, default_delegate_contact (emergency phone), theme_mode, profile_image_uri, remembered_email, biometric_email.

4.3 Formats

- Stored dates: dd/MM/yyyy (pets, expenses, records); completion keys: yyyy-MM-dd; times: hh:mm AM/PM.
- Alarm request codes: task id (routines), pet id + 10000 (vaccinations), 500000 + record id (health records).

5. Non-functional requirements

Area	Requirement
Offline / privacy	Works fully offline except maps and address search. No personal data leaves the device.
Security	Passwords never stored in plain text; biometric data handled only by Android; queries restricted to the logged-in owner; exported activities limited to launcher, import and share.
Performance	Database reads/writes and image decoding run off the main thread; images are downsampled; search is debounced (250 ms).
Reliability	Alarms survive reboot and app updates; failed writes roll back the UI with a message; duplicate imports are prevented.
Usability	Material 3 design, light and dark themes, confirmation before every destructive action, clear error messages, empty states on every list.
Accessibility	Content descriptions on icons, photos, chips and toggles; 40 dp minimum touch targets on day chips; haptic feedback.
Compatibility	Android 10 (API 29) and newer; exact-alarm and biometric fallbacks for older/limited devices.

5.1 Permissions

Permission	Why
POST_NOTIFICATIONS	Show reminders (Android 13+)
SCHEDULE_EXACT_ALARM	Fire reminders at the exact time
RECEIVE_BOOT_COMPLETED	Restore alarms after restart
VIBRATE	Haptic feedback / notification vibration
INTERNET	Map tiles and Nominatim address search
ACCESS_FINE / COARSE_LOCATION	Centre the map on the user (asked only when the my-location button is used)

6. Suggested build order

If you are building the project from this document, this order keeps every step testable:

Phase	Build	Done when
1	Project setup, theme (light/dark), SQLite helper with users table, SessionManager	App launches, theme switch works
2	Welcome / Sign up / Login / Forgot password, password hashing, Remember me, logout	A user can register, log in and out
3	Pets table + Add/Edit pet (validation, species, allergies, photos) + Pet details + delete	Pets can be created, viewed, edited, deleted
4	Tasks + task_completions tables, Add task, TaskSchedule (daily/weekly/monthly), CompletionRules	Unit tests for schedule rules pass
5	Dashboard with bottom tabs, Home page (progress, stats, up next, pet cards), Tasks page, Checklist	Routines can be ticked with all rules
6	Reminders: notification channel, AlarmManager, ReminderReceiver, boot recovery, vaccine reminder	Notifications fire and repeat
7	Health records, vaccination card, HealthSchedule, Notifications screen	Future records create reminders
8	Expenses: add, filters, summary, breakdown, swipe delete, PDF/CSV export	Reports open from Downloads/PetCare export
9	Locations with osmdroid map picker, geocoding, directions; link place to task	Places saved with pins
10	Delegate SMS, Import (.ics/.csv), Share receiver & deep link, Pet history	Importer unit tests pass
11	Profile settings, Personal info, Change password, biometric + Google sign-in, Help/Privacy/Terms, shake	All menu items work
12	Polish: empty states, accessibility, background threading, testing on Android 10 and 14+	Test checklist passes

7. Observations and possible improvements

These points were found while reading the code. They are not requirements, but useful for a project report ("limitations / future work") or as fixes:

- **Forgot password has no verification.** Anyone who knows a registered email can set a new password. An email OTP or security question would fix this.
- **Change password does not ask for the current password.**
- **Password hashing** uses a single SHA-256 round; PBKDF2/bcrypt with a random salt is stronger.
- **"Ends on" date and "Delegate this task"** are saved with a routine but not used anywhere yet (routines never stop repeating).
- **Expense amount on a routine** is stored but not added to the Expenses screen.
- **Notifications screen "Yesterday" section** actually lists today's routines as Done/Missed.
- **Data is device-only:** no cloud sync or backup/restore; uninstalling deletes everything.
- **Currency** is fixed to \$ (US format); a setting for NPR/Rs. would suit local users.
- **Support email** (support@petcare.app) is a placeholder that must be replaced before release.